Infosys_HotelRevAI – Milestone - 4

TOPIC :

AI-Driven Hotel Revenue Analytics and Business Intelligence using Power BI

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Tool Used: Microsoft Power BI

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1. Introduction

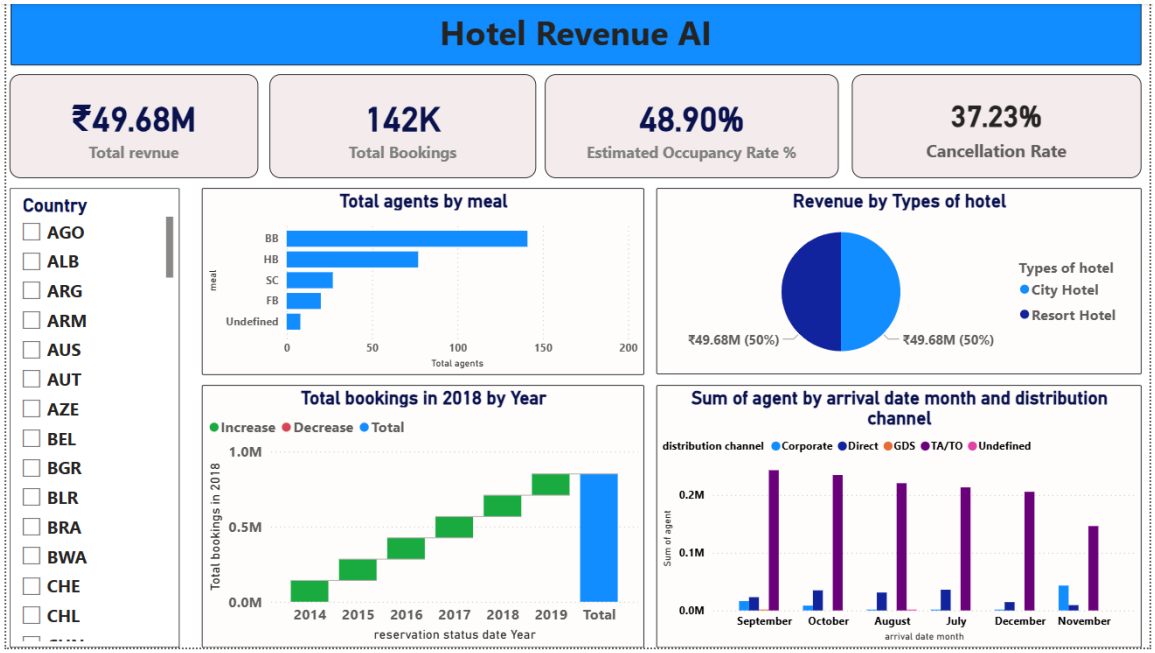
The hospitality industry generates large volumes of booking and customer data every day. However, raw data alone does not help hotel management make effective decisions unless it is properly analyzed and visualized. The Infosys_HotelRevAI project is designed to address this gap by transforming hotel booking data into meaningful insights using Microsoft Power BI.

This project focuses on analyzing hotel revenue, occupancy, customer behavior, and operational performance through interactive dashboards. Advanced analytics techniques such as DAX measures and time-based comparisons are used to provide a clear understanding of business performance.

The main objective of this documentation is to explain the DAX calculations and business metrics used in the project, highlighting their relevance in real-world hotel management scenarios.

2. Project Dashboards Overview

2.1. Hotel Revenue AI (Overview)



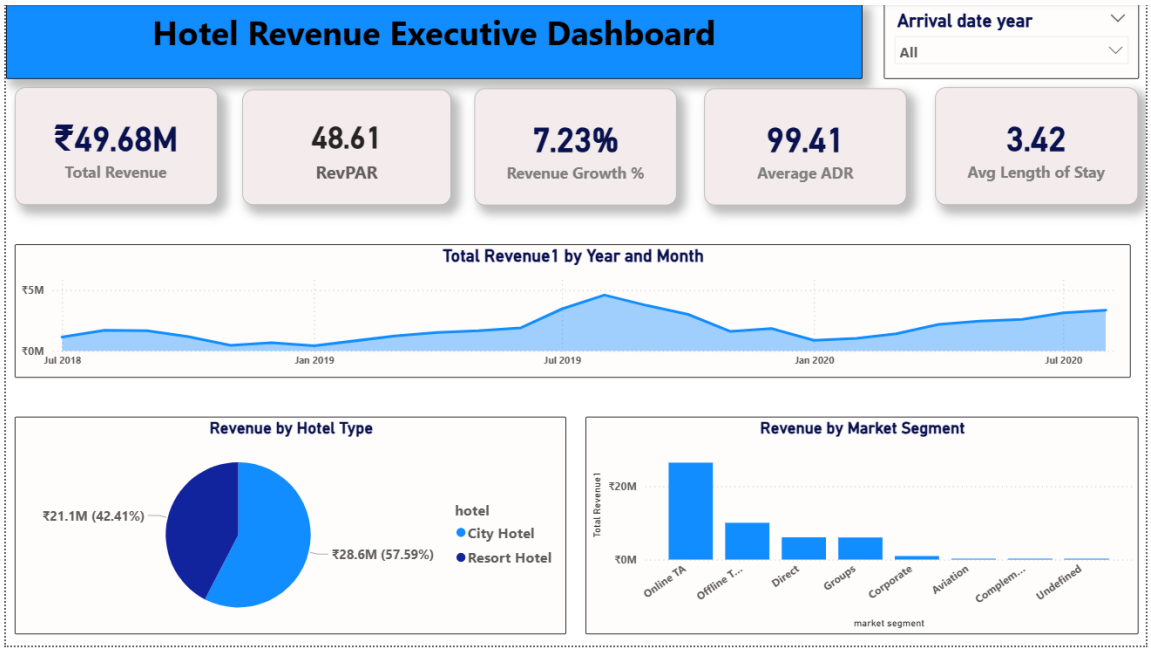
Purpose:
Provides a high-level snapshot of overall hotel performance.

Key Insights:

- Total Revenue
- Total Bookings
- Occupancy Rate
- Cancellation Rate
- Revenue distribution across hotel types

This dashboard is primarily used by management for quick performance monitoring.

2.2. Executive Financial Overview



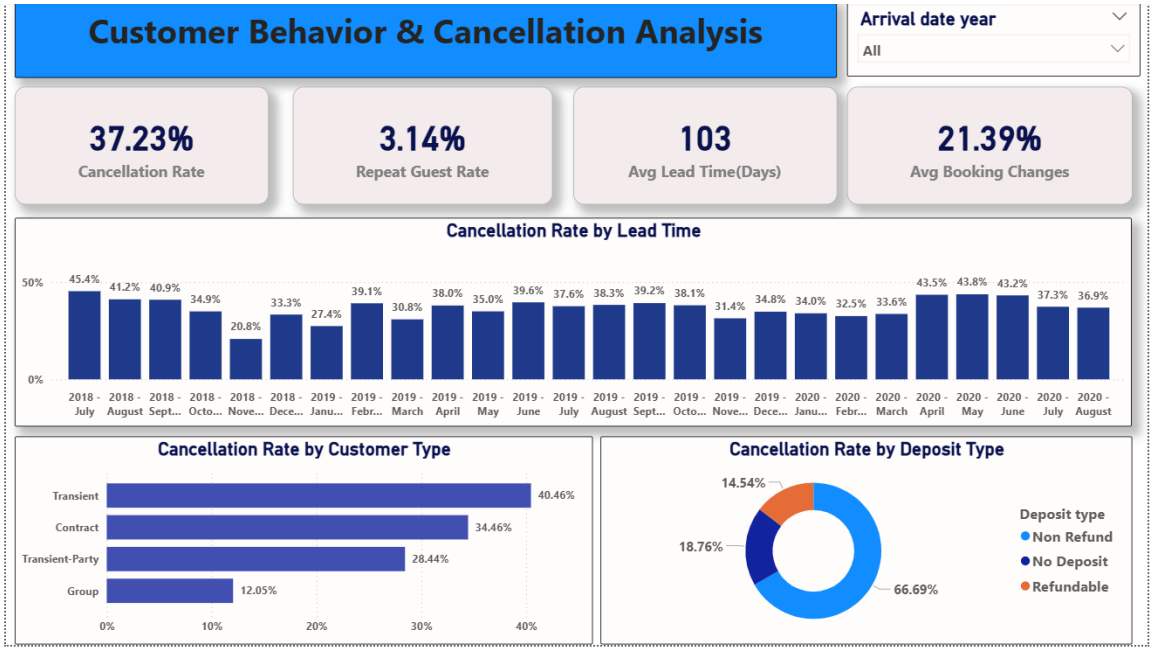
Purpose:
Focuses on strategic revenue and pricing metrics.

Key Insights:

- Revenue Growth %
- RevPAR (Revenue Per Available Room)
- Average Daily Rate (ADR)
- Monthly revenue trends

This dashboard supports executive-level financial planning and performance evaluation.

2.3. Customer & Booking Behavior Analysis



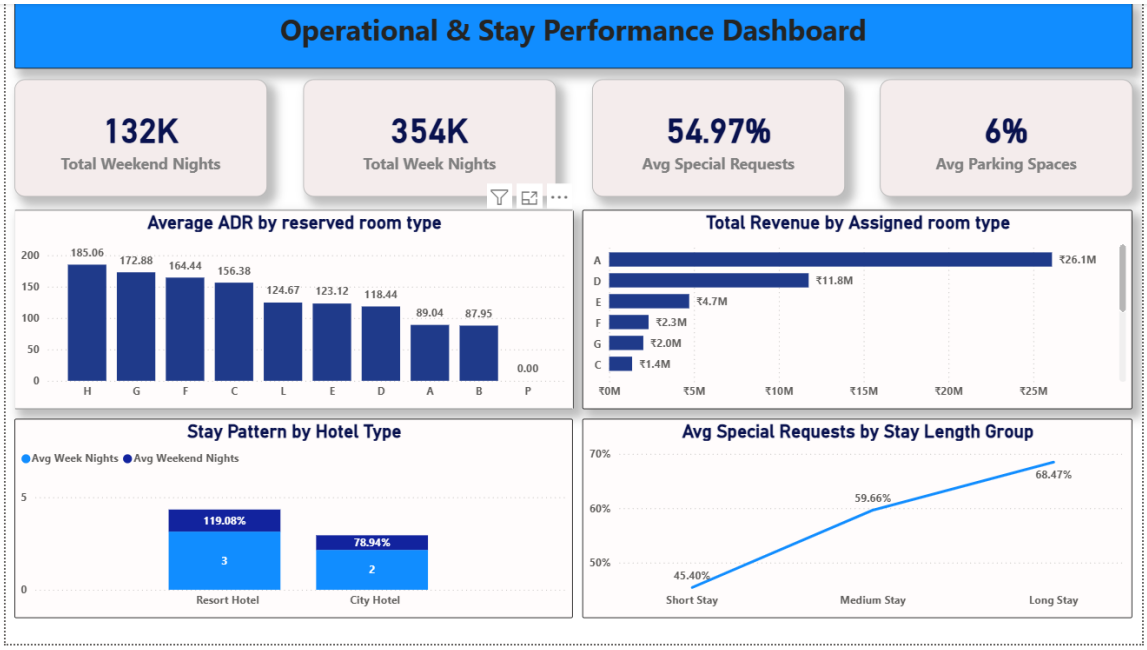
Purpose:
Analyzes customer behavior and booking reliability.

Key Insights:

- Cancellation Rate
- Repeat Guest Rate
- Average Lead Time
- Booking behavior trends

This dashboard helps improve booking policies and customer retention strategies.

2.4. Operational & Stay Performance Dashboard



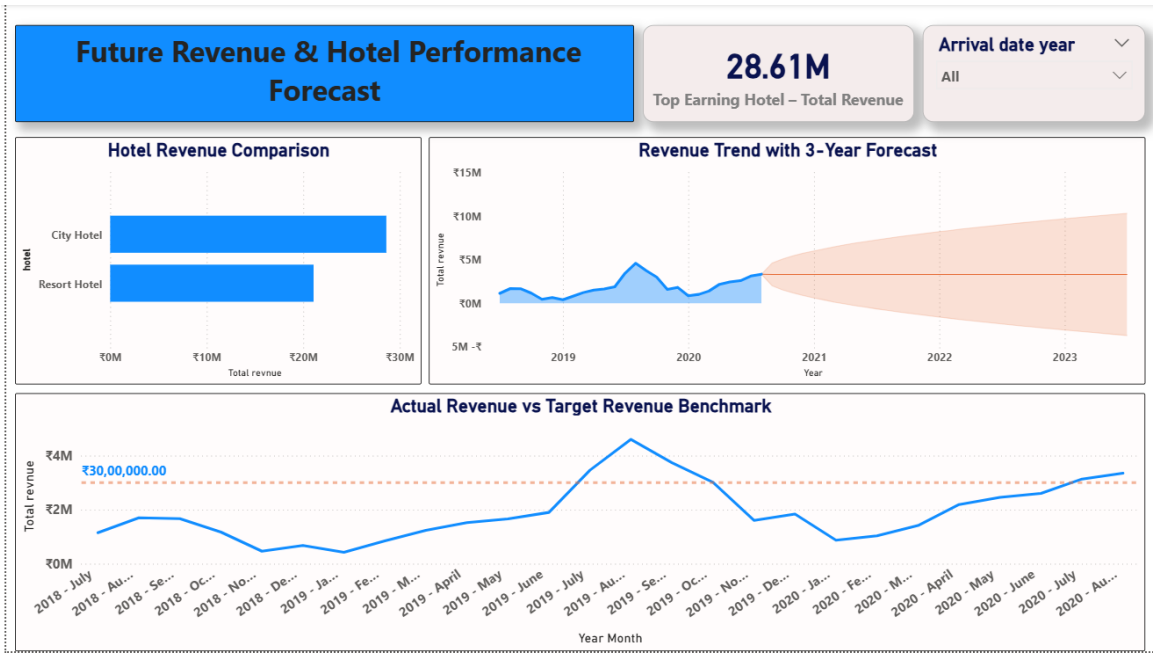
Purpose:
Analyzes operational workload and stay patterns.

Key Insights:

- Weekday vs Weekend stays
- Room utilization
- Special requests
- Parking space usage

This dashboard supports operational planning and resource optimization.

2.5 Future Revenue & Target Performance Dashboard



Purpose:
Focuses on future revenue trends, forecasting, and benchmark comparison.

Key Insights:

- Revenue trend with 3-year forecast
- Actual Revenue vs Target Revenue Benchmark
- Hotel revenue comparison
- Top Earning Hotel KPI
- Expected Revenue (Next 12 Months)

This dashboard supports strategic planning and future decision-making.

3. Core Revenue DAX Measures

DAX measures are the analytical foundation of this project. They allow dynamic calculations that update automatically based on filters and slicers.

3.1 Total Revenue

- Total Revenue is calculated by summing the revenue generated from all bookings. This metric represents the overall financial performance of the hotel.

```
Total Revenue = SUM(Append1[Total_Revenue])
```

3.2 Total Bookings

Business Meaning: Total number of bookings.

```
Total Bookings = COUNT(Append1[reservation_status])
```

3.3 Revenue Growth %

- Revenue Growth Percentage measures the month-over-month change in revenue. It helps management understand whether business performance is improving or declining over time.

```
Revenue Growth % =
VAR CurrentRevenue = [Total Revenue]
VAR PreviousRevenue =
    CALCULATE(
        [Total Revenue],
        DATEADD(Append1[Full Date], -1, MONTH)
    )
RETURN
DIVIDE(CurrentRevenue - PreviousRevenue, PreviousRevenue)
```

4. Operational Performance DAX Measures

4.1 Occupancy Rate %

- Occupancy Rate calculates the proportion of available room nights that are occupied. This metric is critical for evaluating operational efficiency.

```
Occupancy Rate % =
DIVIDE(
    SUM(Append1[stays_in_week_nights]) +
    SUM(Append1[stays_in_weekend_nights]),
```

```
    COUNT(Append1[hotel]) * 7
)
```

4.2 Average Daily Rate (ADR)

- Average Daily Rate (ADR) represents the average price paid per occupied room and is used to analyze pricing strategy.

```
Average ADR = AVERAGE(Append1[adr])
```

4.3 RevPAR

- RevPAR (Revenue Per Available Room) combines ADR and Occupancy Rate to measure revenue efficiency and is considered the most important KPI in hotel analytics.

```
RevPAR = [Average ADR] * [Occupancy Rate %]
```

5. Customer Behavior DAX Measures

5.1 Cancellation Rate %

- Cancellation Rate measures the percentage of bookings that were cancelled, helping identify booking risk and customer reliability.

```
Cancellation Rate % =
DIVIDE(
    [Total Cancelled Bookings],
    [Total Bookings]
)
```

5.2 Average Lead Time

Average number of days between booking and arrival.

- Average Lead Time represents how far in advance customers typically make bookings, which is useful for demand forecasting.

```
Avg Lead Time = AVERAGE(Append1[lead_time])
```

6. Service & Stay Metrics

6.1 Total Week Nights

```
Total Week Nights = SUM(Append1[stays_in_week_nights])
```

6.2 Total Weekend Nights

```
Total Weekend Nights = SUM(Append1[stays_in_weekend_nights])
```

6.3 Total Special Requests

```
Total Special Requests = SUM(Append1[total_of_special_requests])
```

7. BUSINESS METRICS & KPI INTERPRETATION

This section focuses on the business significance of the metrics rather than their calculations. These KPIs help hotel management make informed strategic and operational decisions.

7.1 Strategic Revenue Metrics

- **Total Revenue** represents the overall financial performance of the hotel.
- **Revenue Growth %** indicates whether the hotel's revenue is improving or declining over time.
- **RevPAR** measures how effectively available rooms are generating revenue and is considered a key indicator of hotel profitability.

These metrics are mainly used by **executive management** to assess business health.

7.2 Operational Performance Metrics

- **Occupancy Rate** shows how efficiently hotel rooms are being utilized.
- **Total Week Nights and Weekend Nights** help distinguish between business travel demand and leisure travel demand.
- Higher weekday stays usually indicate corporate bookings, while higher weekend stays reflect leisure tourism.

These metrics support decisions related to **staffing, pricing, and capacity planning**.

7.3 Customer Behavior Metrics

- **Cancellation Rate** helps identify booking risk and reliability.
- **Average Lead Time** shows how early customers plan their bookings.

- **Repeat Guest Rate** reflects customer loyalty and satisfaction.

Understanding these metrics allows hotels to improve **booking policies and marketing strategies**.

7.4 Service Usage Metrics

- **Special Requests** indicate guest service expectations.
- **Parking Spaces Usage** reflects facility demand and infrastructure utilization.

Higher values suggest the need for enhanced service management and operational readiness.

8. Future Revenue & AI-Driven Forecasting KPIs

8.1 Top Earning Hotel – Total Revenue

```
Top Earning Hotel - Total Revenue =

MAXX(

    VALUES(Append1[hotel]),

    [Total Revenue]

)
```

Business Value:
Identifies the strongest revenue contributor.

8.2 Expected Revenue (Next 12 Months)

```
Expected Revenue (Next 12 Months) =
CALCULATE(
    [Total Revenue],
    DATESINPERIOD(
        Append1[Full Date],
        MAX(Append1[Full Date]),
        12,
        MONTH ))
```

Business Value:
Provides short-term revenue outlook, critical for executive planning.

9. Data Visualization

Data visualization is the graphical representation of data using charts, graphs, and dashboards. Instead of analyzing large tables of numbers, visual representations allow users to quickly understand important insights from the data. By presenting information visually, complex datasets become easier to interpret and analyze.

Visualization helps highlight trends, patterns, and relationships that may not be easily visible in raw data. It plays a very important role in business intelligence systems where organizations rely on visual dashboards to monitor performance and make informed decisions.

One of the main advantages of data visualization is that it helps users understand large amounts of information quickly. Complex data becomes easier to interpret, allowing analysts and decision makers to identify important trends and patterns in the dataset. Visualization also improves communication of insights to stakeholders, as charts and dashboards present information in a simple and clear format.

Another important benefit is that it supports faster and more accurate decision making. Instead of manually analyzing spreadsheets and reports, managers can view summarized insights through visual dashboards. This reduces the time required for manual data analysis and increases overall efficiency.

Various visual tools are used in data analytics to represent different types of information. Bar charts are commonly used to compare values across categories. Line charts are useful for identifying trends and patterns over time. Pie charts represent proportions and help visualize percentage distribution. KPI cards display important business metrics such as revenue, bookings, and growth rates. Dashboards combine multiple visualizations in a single interface, providing a complete overview of business performance.

In this project, Microsoft Power BI dashboards are used to visualize hotel revenue, customer behavior, operational performance, and forecasting insights. These interactive dashboards allow users to explore data through filters and slicers, enabling deeper analysis of hotel business performance.

10.Data Storytelling

Data storytelling is the process of explaining insights using a combination of data, visualizations, and descriptive explanations. It transforms raw data into meaningful information that people can easily understand.

In business analytics, data storytelling plays an important role because it helps decision makers understand the meaning behind the data. Instead of presenting only charts or numbers, storytelling combines visuals with explanations to clearly communicate insights and business implications.

Through data storytelling, complex analytical results are converted into understandable narratives. This makes it easier for stakeholders, managers, and executives to interpret the data and take appropriate actions.

In the Hotel Revenue AI project, dashboards act as storytelling tools. Each dashboard focuses on a specific aspect of hotel performance, such as revenue analysis, customer booking behavior, operational efficiency, and future revenue forecasting. By presenting data in a structured visual format, the dashboards guide users through the insights and help them understand the overall business performance.

Effective data storytelling improves communication of analytical results and ensures that insights derived from the data can support strategic and operational decision making.

11. Artificial Intelligence in Data Analytics

Artificial Intelligence (AI) enables machines to perform tasks that normally require human intelligence. In data analytics, AI is widely used to analyze large datasets, identify hidden patterns, and generate predictive insights.

AI-powered analytics tools can automatically process large volumes of data and detect trends that might not be easily visible through traditional analysis. This helps organizations make more informed and data-driven decisions.

One of the most important applications of AI in business intelligence is predictive analytics. Predictive analytics uses historical data to forecast future trends and outcomes. In the hospitality industry, AI can help predict hotel revenue, customer demand, seasonal booking patterns, and operational requirements.

In this project, forecasting techniques are used to estimate future revenue trends for hotels. These predictive insights help hotel management plan pricing strategies, optimize resource allocation, and prepare for future demand.

Another advantage of AI in data analytics is that it reduces manual effort in analyzing complex datasets. Automated analysis saves time and improves accuracy, allowing analysts to focus more on strategic decision making rather than manual calculations.

By integrating data visualization, business intelligence, and AI-driven forecasting, the Hotel Revenue AI project demonstrates how modern analytics solutions can support effective decision making in the hospitality industry.

12. Conclusion

The Infosys_HotelRevAI project demonstrates how business intelligence tools can be used to transform raw hotel booking data into meaningful insights. By using Microsoft Power BI, the project successfully analyzes key aspects of hotel performance such as revenue generation, customer booking behavior, cancellation trends, and operational efficiency.

Through the use of interactive dashboards and DAX calculations, the project provides a clear understanding of important business metrics including Total Revenue, Occupancy Rate, RevPAR, Average Daily Rate, and Cancellation Rate. These metrics help hotel management evaluate business performance and identify areas that require improvement.

The dashboards developed in this project allow decision makers to monitor hotel performance from multiple perspectives, including financial performance, customer behavior, operational workload, and service utilization. This helps management make informed decisions related to pricing strategies, customer retention, and resource planning.

In addition, the inclusion of revenue forecasting and benchmark comparison enhances the analytical capabilities of the system by providing insights into potential future trends. This supports strategic planning and enables hotels to prepare for future demand and revenue opportunities.

Overall, this project demonstrates how data visualization, business intelligence, and analytics techniques can support data-driven decision making in the hospitality industry. By converting complex data into clear and interactive insights, the Hotel Revenue AI system helps improve operational efficiency, strategic planning, and overall business performance.